Features: Useful Clues

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Features: Useful Clues

> Prepare Features: Useful Clues: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Choose recorded properties that preserve distinctions needed for a stated task

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-1/8, ai-2/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a category definition using explicit **evidence** and boundary rules

Reference example, not the only acceptable wording: In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room.

Equality at 30 is accepted.

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and reconstruct a small image using an explicit grid convention

Reference example, not the only acceptable wording: The upper row has black, white, black; the lower row has white, black, white. Reading back gives the supplied two rows.

There are three black pixels among six total positions.

Input, Process, Output

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

Look for: Trace information through a system without confusing information with a physical action

Reference example, not the only acceptable wording: **Input** includes both request fields and all three catalogue records. The **process** checks science AND pages<=120.

Output contains A only; B fails the length condition and C fails the genre condition. Listing A does not establish that a copy was reserved.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

Concept to explain

Some observable clues help a decision more than others.

Term	Meaning
feature	represented clue used in a decision
relevant	connected to the task goal
distraction	available clue that should not drive the decision
spurious clue	accidental pattern that may fail later

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Students investigate representation before **fitting** a model. Keep the paper rule explicit: positive known measurements, fixed orientation, no thickness in the card task, and both boundaries inclusive.

Length and width suffice for that limited rule. A collision between **feature record**s with different correct labels demonstrates lost information; it does not prove that every task needs these same **input**s.

Explain **leakage** through timing and the copied sticker, without requiring model **training** or statistics.

Readiness answer

Check the card fields against the same explicit policy and version. Agreement alone can preserve a shared mistake; it does not prove that an event exists or has permission.

Four cards matching the worked table, with dimensions, colour and identifying letters written as text

Two paper columns headed Before the decision and After inspection, plus removable label stickers

Blank records headed Length in cm and Width in cm, and the independent display-tray task from W4

End goal for the pictorial case

By the end, I can choose task-**relevant feature**s and avoid a clue created after the decision.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Features: Useful Clues using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT measured candidate clues

2. Process — the deciding step

PROCESS compare useful patterns and ignore **distraction**s

3. Output — the result

OUTPUT predicted cat/dog label with **evidence**

Key idea

CORE IDEA Some observable clues help a decision more than others.

How the mechanism works

UNDER THE HOOD A **feature** is a represented property used by a decision **process**. Predictive usefulness depends on the task and data.

Spurious **feature**s correlate in a sample without reflecting the intended concept and often fail when conditions change.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE Ear shape, muzzle length, and paw shape may help distinguish cats from dogs; background color and photo border usually do not.

Step 1

STEP 1 — OBSERVE measured candidate clues

Step 2

STEP 2 — TRACE compare useful patterns and ignore **distraction**s

Step 3

STEP 3 — CONCLUDE predicted cat/dog label with **evidence**

Model answer

MODEL REASONING Useful **feature**s describe the animal and remain available across settings. Borders and backgrounds describe the photo setup.

Even ear shape may fail for unusual breeds or blocked views, so multiple clues and uncertainty are needed.

A claim to avoid

DO NOT OVERCLAIM A clue can be observable yet still be ir**relevant**, unfair, or fragile.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

The teaching case

Choose the **relevant feature** and explain why size is insufficient.

Exact explanation to model

Why the result follows

Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

Explain the added worked example

Explain the added worked example · 1

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number.

The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality.

Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

Explain the added worked example · 2

A **feature** is a recorded **input** property, such as width in centimetres. The **target label** is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the **input**s.

A card number can help us find a record, but knowing its number does not tell us its dimensions.

Explain the added worked example · 3

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a **learned model** would need.

Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

Explain the added worked example · 4

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same **input**, 10, is paired with two different correct answers.

It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that **input** alone.

A **feature record** needs a consistent order and units. Here we write length in centimetres first, then width in centimetres.

The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

Key terms

Term	Meaning
Feature	A recorded property supplied as an input to a system
Target label	The answer a supervised learning task asks a model to predict
Feature record	The selected input values for one example, in a defined order
Relevant feature	An input property that helps make distinctions needed for the stated task
Leakage	Information used during development that would not legitimately be available when making the intended prediction

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Card measurements and colour	Label under the guide	What the comparison shows
10 by 7 cm and blue	Fits	Compare with the next card
10 by 9 cm and blue	Too large	Same length and colour but width changes the label
13 by 7 cm and blue	Too large	Same width and colour as the first card but length changes the label
10 by 7 cm and red	Fits	Changing only colour does not change this guide

Model the reasoning aloud · 5

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule.

That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Model the reasoning aloud · 6

Suppose every **fitting** card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the shortcut.

Check a proposed **feature** on deliberately different cases before trusting the practice score.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Features: Useful Clues

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

The facts we will use

> **Supplied facts**
>
> Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

Supplied facts

A green acceptance sticker is attached only after the shelf check.

AI ACADEMY / AI-1/8

Dimensions determine shelf acceptance

	Width cm	Height cm	Outcome
P stripes	6	4	fits
Q stripes	7	4	width fails
R dots	6	5	height fails
S dots	6	4	fits

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/8

Features: Useful Clues

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Width check

P,Q,R,S have widths 6,7,6,6.

2. Height check

Their heights are 4,4,5,4.

3. Apply both limits

P and S fit; Q fails width; R fails height.

4. Reject a shortcut

The later sticker cannot be an input available before checking.

CHANGE ONE THING

Change Q's width to 6 without changing its height.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Width check	P,Q,R,S have widths 6,7,6,6.
2. Height check	Their heights are 4,4,5,4.
3. Apply both limits	P and S fit; Q fails width; R fails height.
4. Reject a shortcut	The later sticker cannot be an input available before checking.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Comparing P with Q isolates width; comparing P with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce.

What this example does not establish

Keep the stated limit

These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

Change one thing in the pictured case

Change one thing

Change: Change Q's width to 6 without changing its height.

Predicted result

Prediction: Q fits the shelf.

Why the result follows

Reason: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Features: Useful Clues, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

2. State the decision to explain

Write the question to explain: Choose the **relevant feature** and explain why size is insufficient.

3. Reveal the worked reasoning

Reveal after **prediction**s: Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

Hold back the independent task

Keep for an independent attempt: A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful **feature** and a distracting **feature**. Test your choice on a blue circle.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark **relevant** details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Length 12 cm is usable before inspection and passes its criterion. The unreadable width remains unknown, so it might be at most 8 or above 8; the available measurements do not establish fits or too large.

The final sticker does not exist before inspection, so it cannot supply an **input** at the time the answer is needed. Using it would leak the later answer.

The exit answer may use red 10 by 7 cm, which fits, or blue 13 by 7 cm, which is too large. Check the dimensions and use new cases that break the practice colour pattern.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve Week 7: agreement is not proof of a correct label. State the new target and check both inclusive boundaries.

Stage	Teacher action
7 to 20 minutes	Show cards A and B. Cover their widths and identifying letters. Ask what identical length-only inputs allow the system to distinguish, then reveal width.
20 to 33 minutes	Trace the four-row comparison table. Have students cite the pair showing each measurement matters and the pair showing colour does not enter this rule.
33 to 45 minutes	Complete W1 and W2 with partners. Require a constructed pair and reasons; do not accept only a list of feature names.
Session 2 start 0 to 8 minutes	Retrieve the same-input problem using length 11 width 5 and length 11 width 10. Both length-only records are 11; the first fits and the second is too large.
8 to 23 minutes	Use W3 to break the colour shortcut. Place the sticker after the decision line and discuss why its apparent predictive success is misleading.
23 to 38 minutes	Collect W4 independently. Look for height and thickness, a valid equal-height pair, boundary reasoning and the timing explanation.
38 to 45 minutes	Use W5 and the exit response to revisit task-dependent relevance. Preview algorithms that update a recorded state one step at a time.

Reduce help deliberately

Completion practice

Completion task: Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels.

Explain why a system receiving only width cannot always label both correctly.

Reasoning to listen for

Teacher check: One valid pair is length 11 width 6, which fits, and length 13 width 6, which is too large. Both give the width-only **input** 6 but require different labels.

Without length or other permitted distinguishing information, the system cannot guarantee correct labels for both. Accept any valid pair with positive measurements, equal widths at most 8, and lengths on opposite sides of 12.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Cover width and all identifiers for two same-length cards with different labels. Ask which supplied value tells them apart. Guessing or using typical values cannot guarantee correctness for both.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the **relevant** step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

Reasoned teacher answer

P and S test pattern. The sticker is attached after the decision, so it is unavailable when a new box must be checked.

Changed-case reasoning: Q fits the shelf. Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Features: Useful Clues investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Choose three clues for cat/dog sorting; reject **distraction**s.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR The brightest **feature** is always the best clue.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A stronger model can always recover an omitted measurement: Cover width and all identifiers for two same-length cards with different labels. Ask which supplied value tells them apart.

Guessing or using typical values cannot guarantee correctness for both.

A **feature** that matches practice labels must be useful: Use a red **fitting** card or a blue oversized card. Ask whether the **feature** preserves the stated distinction when the practice pattern changes.

The final label is the best **input feature**: Place the sticker after inspection on the timeline. It is a target for checking or teaching, not an available **input** for a **prediction** before inspection.

Useful **feature**s stay useful for every task: Change the target to red or blue bins. Identical dimensions can now require different answers, so colour matters.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Features: Useful Clues questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Edge shape directly addresses the target. Size can vary within both categories. **Feature** usefulness depends on the task.

Independent question

Independent question · withhold the key

A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful **feature** and a distracting **feature**. Test your choice on a blue circle.

Expected independent answer

Teacher key · Expected independent answer

Shape is useful; colour is distracting for this task. A blue circle is round. A **feature** is useful because it relates to the intended category, not because it is easy to see.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The brightest **feature** is always the best clue.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> Test a large smooth leaf and a small toothed leaf with reliable labels. An edge-based rule predicts smooth and toothed respectively; a size shortcut reverses them.
>
> Two tests expose this specific shortcut, but do not establish performance on every leaf.

Explanations for the additional workbook examples

W1 Separate inputs from the target

Workbook question · W1

For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality.

Choose the measurements needed as **input**s from length, width, colour, card number and the final inspection sticker. State the **target label** and write the **feature record** for a 12 cm by 8 cm red card using a defined order.

Teacher reasoning · W1

Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

W2 Prove that one measurement is insufficient

Workbook question · W2

Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

Teacher reasoning · W2

One valid pair is length 11 width 6, which fits, and length 13 width 6, which is too large. Both give the width-only **input** 6 but require different labels.

Without length or other permitted distinguishing information, the system cannot guarantee correct labels for both. Accept any valid pair with positive measurements, equal widths at most 8, and lengths on opposite sides of 12.

W3 Challenge an apparent colour shortcut

Workbook question · W3

A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels.

Explain why the perfect practice score did not establish that colour was a reliable **feature** for this task.

Teacher reasoning · W3

A red 10 by 7 cm card fits but the helper says too large; alternatively a blue 13 by 7 cm card is too large but the helper says fits. The practice set paired colour with size and did not test cases that separate them.

The stated guide depends on measurements, so compare **prediction**s with that guide on cases where colour and size no longer follow the shortcut.

W4 Design a record for a new task

Workbook question · W4

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

Teacher reasoning · W4

Record height in cm and thickness in cm. A valid pair is height 8 thickness 1, which passes, and height 8 thickness 3, which fails: equal height hides the required thickness distinction.

The 9 cm high, 2 cm thick tile passes both inclusive boundaries. Pattern name does not affect the stated rule. The pass sticker is created after checking and copies the target; using it to predict before checking is **leakage**.

W5 Notice when the task changes

Workbook question · W5

The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded **feature** becomes useful? Explain why length and width alone may now be insufficient.

Give two cards that support your explanation.

Teacher reasoning · W5

Colour becomes useful because the new target is red or blue. A red 10 by 7 card and a blue 10 by 7 card have identical length-width records but require different bins.

Feature relevance depends on the task; reusing the previous **feature** set without reviewing the target can discard the needed distinction.

Assess independent transfer

Independent question · withhold the key

Selected case: A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed.

Before checking, you can record height, thickness and pattern name. A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

Selected case answer

Teacher key · Selected case answer

Record height in cm and thickness in cm. A valid pair is height 8 thickness 1, which passes, and height 8 thickness 3, which fails: equal height hides the required thickness distinction.

The 9 cm high, 2 cm thick tile passes both inclusive boundaries. Pattern name does not affect the stated rule. The pass sticker is created after checking and copies the target; using it to predict before checking is **leakage**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the **relevant** skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Features: Useful Clues to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Select the smallest useful **feature**s for lost-and-found sorting and reject clues tied to a particular classroom or owner.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 9: Algorithms as Recipes. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Let students physically cover unselected columns and compare two records side by side. Provide the sentence frame Same **input** ___ but labels ___ and ___ because ___. Use written colour names so no task depends on colour vision.

Extension: ask learners to prove the width-only **input** fails using a second valid pair, or explain how permitting card rotation would change the decision procedure even when the recorded dimensions stay the same.

W4 shows secure independent understanding when the learner selects height and thickness with units, supplies an equal-height pair on opposite sides of the thickness limit, labels the boundary tile correctly, and explains the sticker timing.

Naming **input**s without a counterexample is partial **evidence**. After support, retry with a new tray limit of height 7 cm and thickness 3 cm; ask for a fresh equal-height pair before confirming mastery.

Week 9 introduces algorithms and changing state. Students will trace ordered instructions and record how values change, keeping the initial conditions and the sequence visible.

Plan the delayed return

Suggested return points in this level: ai-1/9, ai-1/11, ai-1/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Features: Useful Clues lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Choose recorded properties that preserve distinctions needed for a stated task

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels.

Explain why a system receiving only width cannot always label both correctly.

Supported practice answer

Reasoned practice answer: One valid pair is length 11 width 6, which fits, and length 13 width 6, which is too large. Both give the width-only **input** 6 but require different labels.

Without length or other permitted distinguishing information, the system cannot guarantee correct labels for both. Accept any valid pair with positive measurements, equal widths at most 8, and lengths on opposite sides of 12.

Check every learner before discussion: Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Cover width and all identifiers for two same-length cards with different labels. Ask which supplied value tells them apart. Guessing or using typical values cannot guarantee correctness for both.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change Q’s width to 6 without changing its height. Require a **prediction** and a reason before comparing. Expected result: Q fits the shelf.

Reason: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

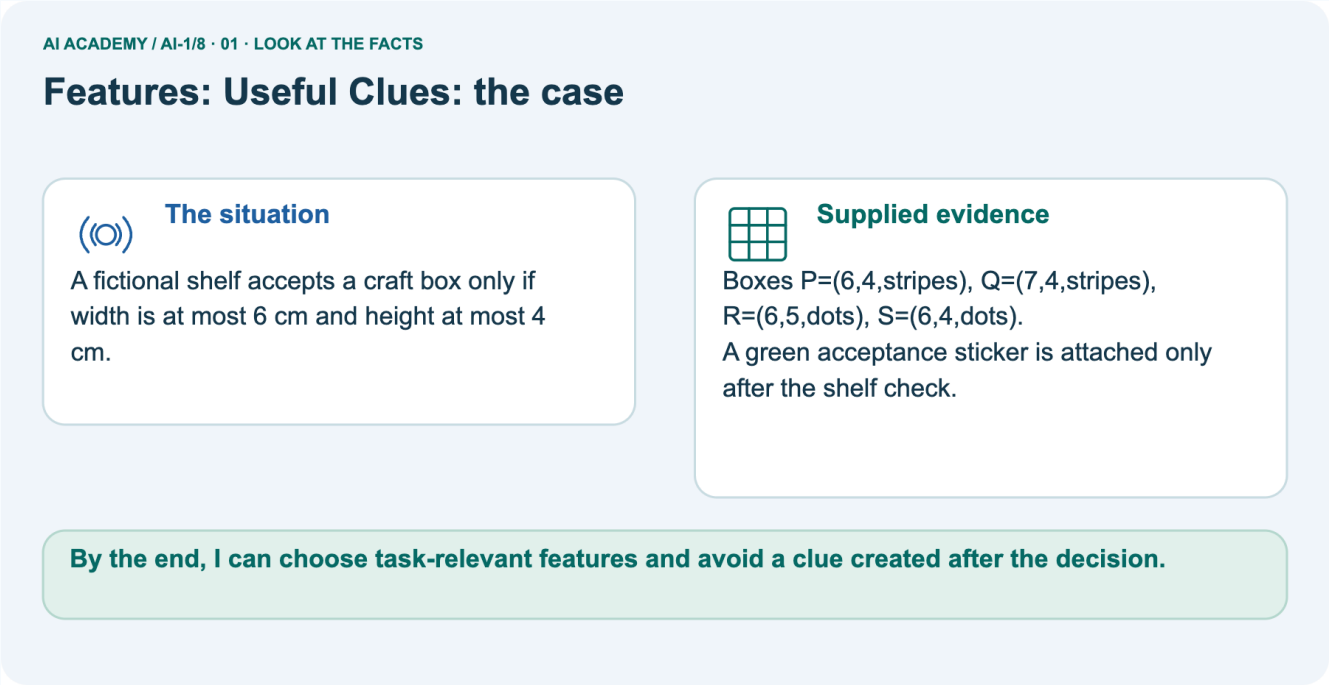
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Features: Useful Clues graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Features: Useful Clues: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm. Supplied facts: Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

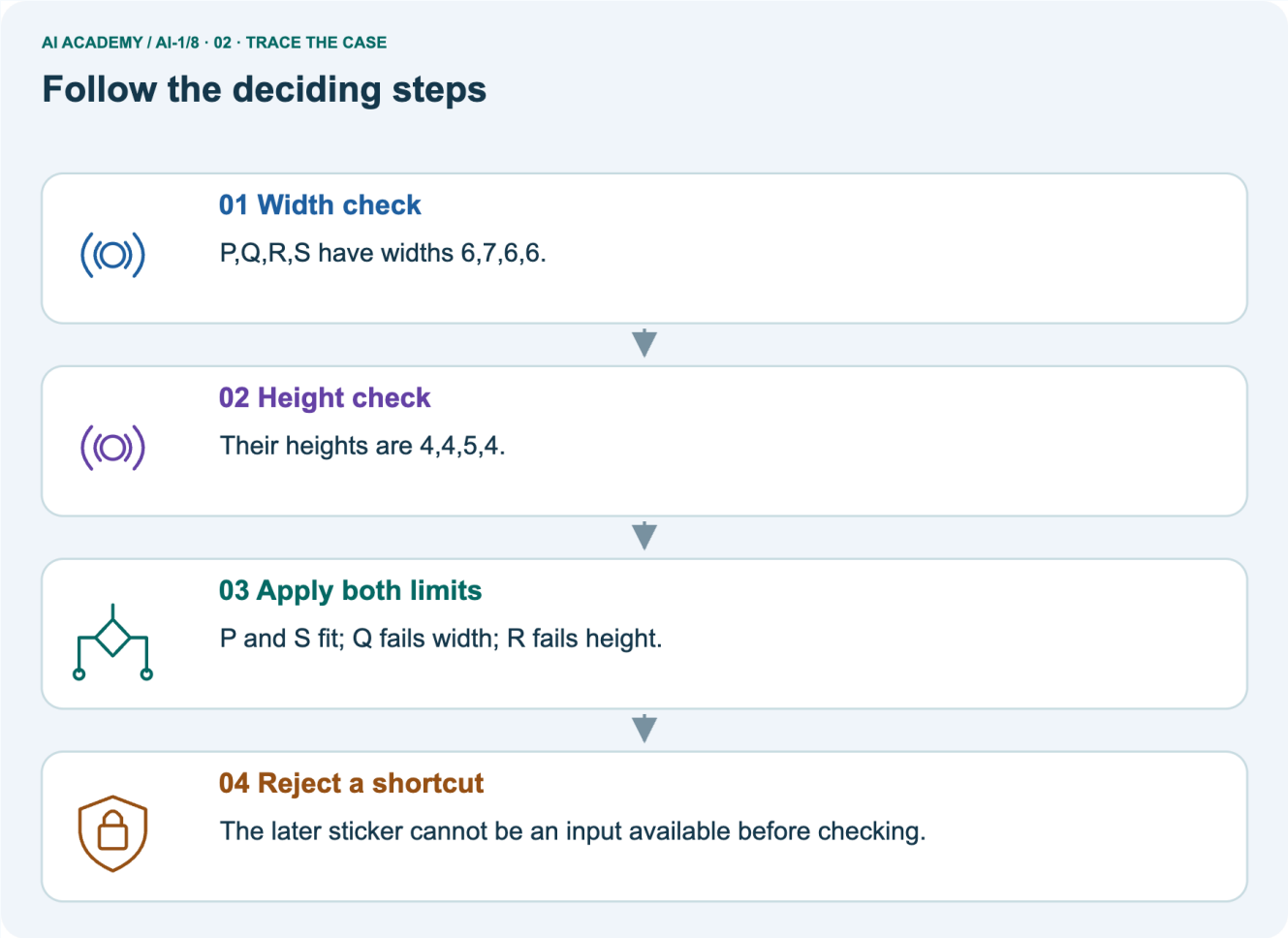
A green acceptance sticker is attached only after the shelf check.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

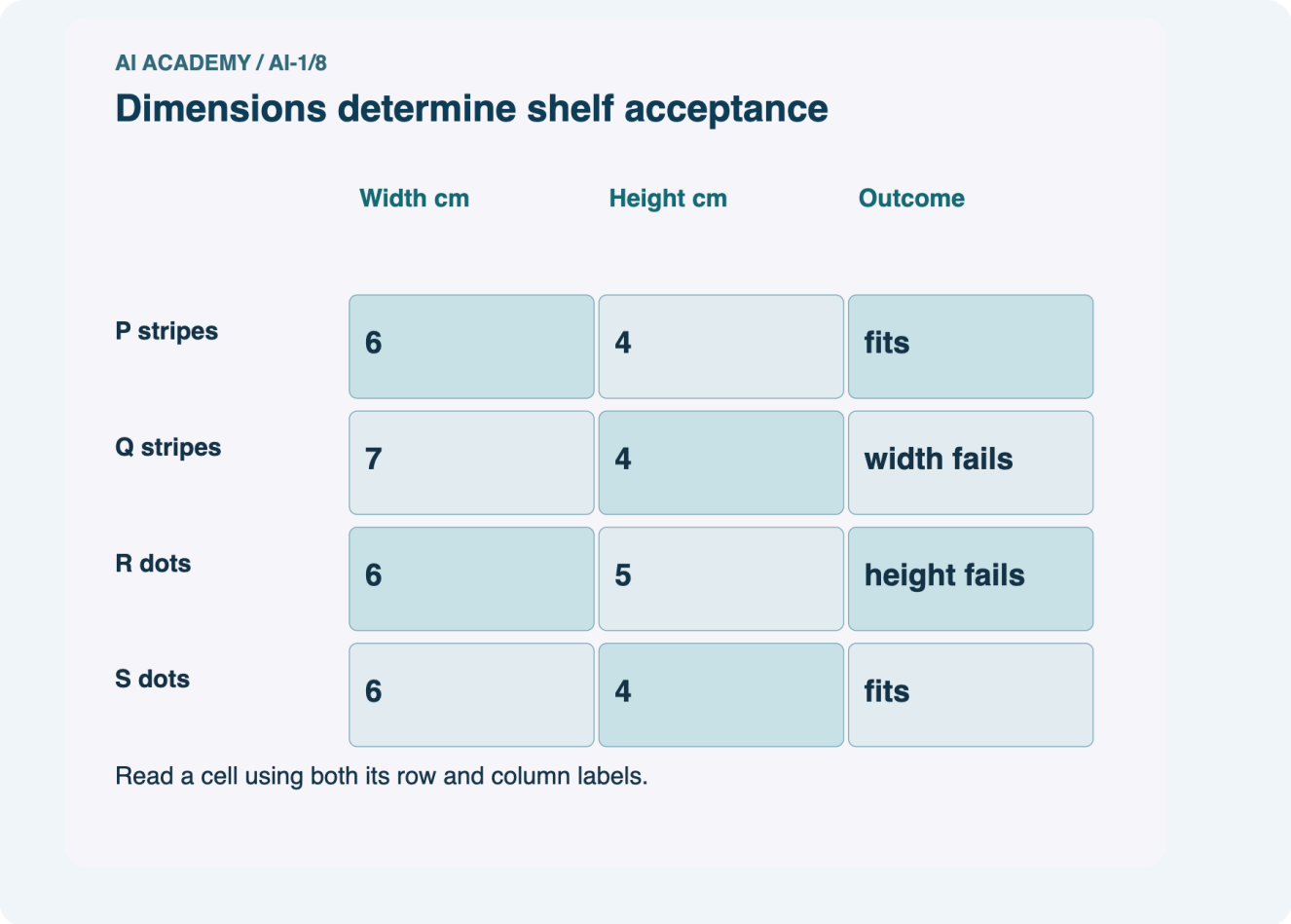
Width check: P,Q,R,S have widths 6,7,6,6. → Height check: Their heights are 4,4,5,4. → Apply both limits: P and S fit; Q fails width; R fails height. → Reject a shortcut: The later sticker cannot be an **input** available before checking.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Dimensions determine shelf acceptance



Follow the labelled picture with the matching explanation. Open the vector diagram

Dimensions determine shelf acceptance. P stripes: Width cm = 6, Height cm = 4, Outcome = fits. Q stripes: Width cm = 7, Height cm = 4, Outcome = width fails. R dots: Width cm = 6, Height cm = 5, Outcome = height fails.

S dots: Width cm = 6, Height cm = 4, Outcome = fits.

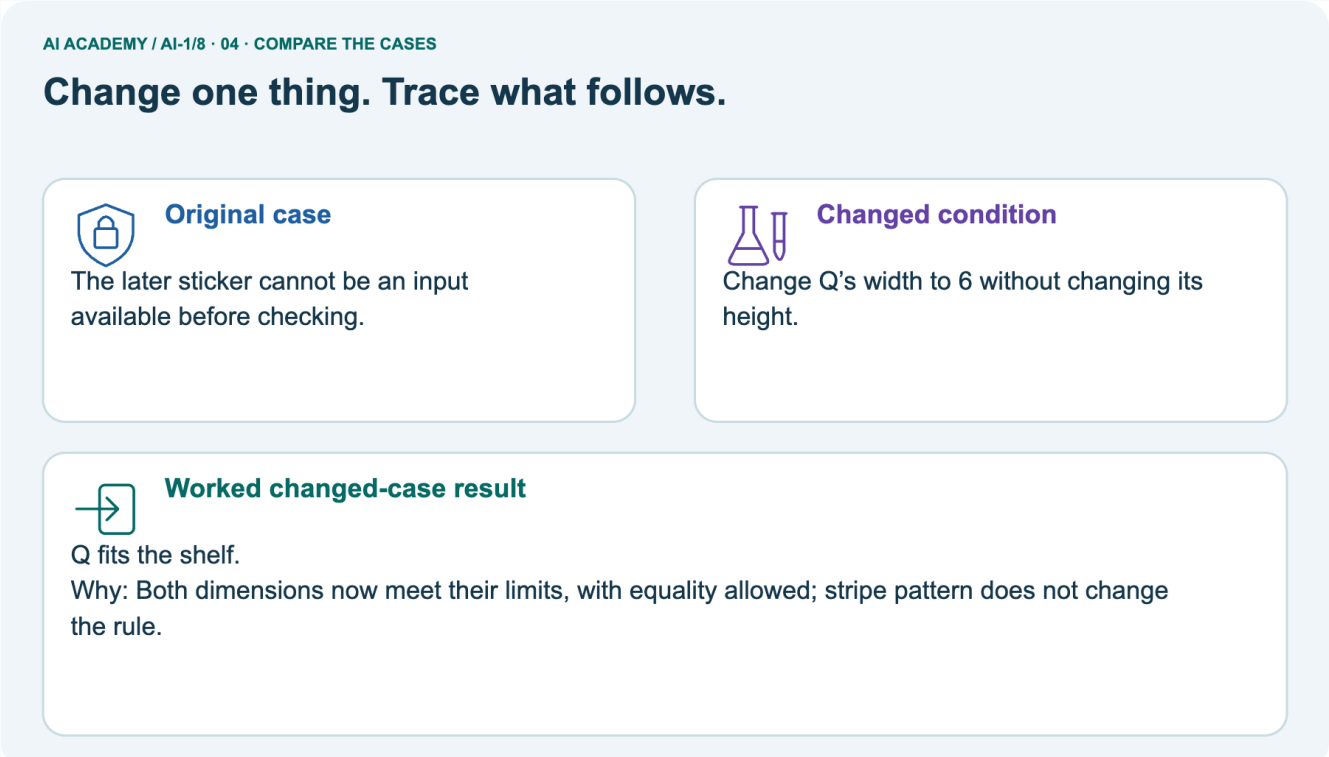
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The later sticker cannot be an **input** available before checking. Changed condition: Change Q's width to 6 without changing its height. Expected result: Q fits the shelf.

Reason: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

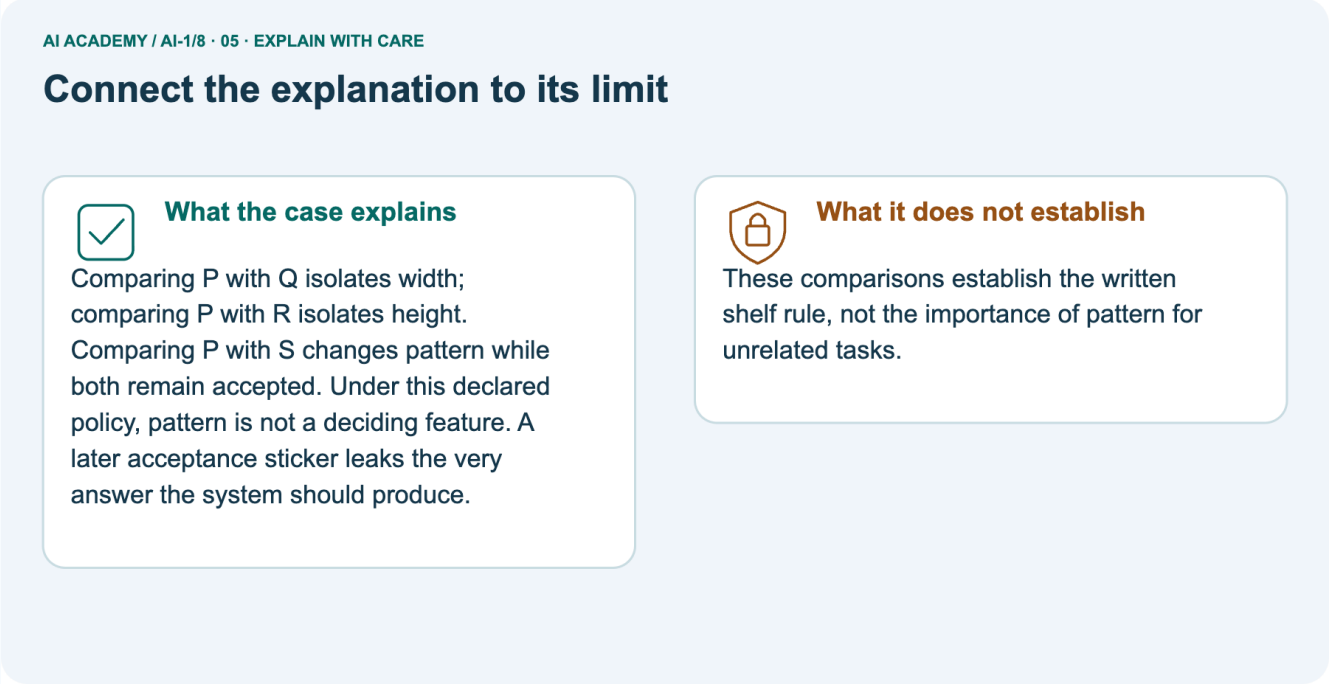
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Both dimensions now meet their limits, with equality allowed; stripe pattern does not change the rule.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Comparing P with Q isolates width; comparing P with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding **feature**.

A later acceptance sticker leaks the very answer the system should produce. Limit: These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

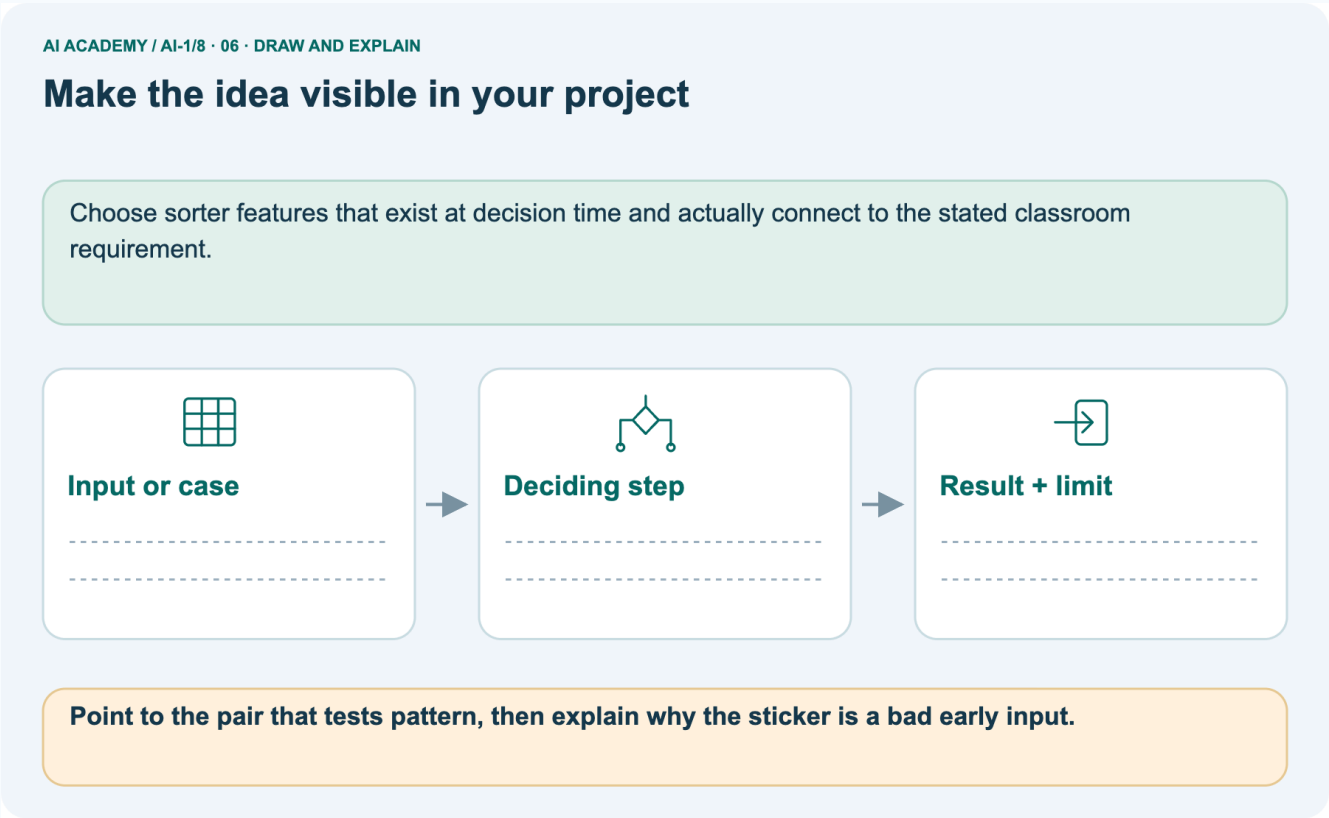
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These comparisons establish the written shelf rule, not the importance of pattern for unrelated tasks.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **feature**s that exist at decision time and actually connect to the stated classroom requirement. Draw your **input**, deciding step, result and limit.

Supported question: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: P and S test pattern. The sticker is attached after the decision, so it is unavailable when a new box must be checked.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.